POSITIVITY OF THE QUARTIC SOLUTION OF A DISCRETE PAINLEVÉ-TYPE RECURRENCE

ABSTRACT. We prove positivity of the solution of the quartic discrete Painlevé-type recurrence arising from the quantum minimal surface $w_2 = w_1^3$. The initial data are given by a quadratic combination of Bessel functions of orders $\pm\frac{1}{4}$. The proof converts the Bessel amplitude into a positive Laplace integral and uses two quartic source variables to reduce the four residue classes modulo 4 to one family of bi-moment matrices. A triangular factorization and Andréief's identity fix the strict sign of every leading principal minor in the four symmetry blocks and give an explicit positive multiple-integral formula for each minor once its predicted sign is divided out. Consequently all signed orthogonal-polynomial norms are positive, and their scaled consecutive ratios give $v_n(a) > 0$ for every $a > 0$ and $n \geq 0$.

1. INTRODUCTION

Discrete Painlevé-type recurrences occur in matrix regularizations of minimal surfaces and in the recurrence coefficients of orthogonal polynomials. The operator formulation of quantum minimal surfaces has its origins in [5] and was developed further in [2], where higher recurrences are discussed. The quartic recurrence below is written explicitly in [10, equation (38)] and in [4]; those works also treat the special-function positive solution in the $d = 3$, or cubic-potential, case.

For the quantum minimal surface associated with the holomorphic curve $w_2 = w_1^{d-1}$, Felder and Hoppe [7] constructed an indefinite Hermitian form and showed that, whenever the relevant leading Gram matrices are nondegenerate, its orthogonal-polynomial norm ratios obey

$$V_n + \frac{1}{a^2} \sum_{j=0}^{d-2} \prod_{k=0}^{d-2} V_{n+j-k} = \frac{n+1}{a}. \tag{1.1}$$

For $d = 4$, after setting $v_n = V_n/a$ and $\epsilon = a^{-2}$, the recurrence becomes

$$v_n \left(1 + v_{n-1}v_{n-2} + v_{n+1}v_{n-1} + v_{n+2}v_{n+1}\right) = (n+1)\epsilon, \qquad n \geq 0, \tag{1.2}$$

where $v_n = 0$ for $n < 0$. The two initial values are determined by

$$v_0 = -\frac{1}{a}\frac{h'(a)}{h(a)}, \qquad v_1 = -\frac{1}{a}\frac{h''(a)}{h'(a)}, \tag{1.3}$$

with

$$h(a) = \frac{\pi^2 a}{4}\Bigg[J_{-1/4}\left(\frac{a^2}{4}\right)^2 - \sqrt{2}\, J_{-1/4}\left(\frac{a^2}{4}\right) J_{1/4}\left(\frac{a^2}{4}\right) + J_{1/4}\left(\frac{a^2}{4}\right)^2\Bigg]. \tag{1.4}$$

Felder and Hoppe proved the required Gram nondegeneracy for $d = 1, 2, 3$ and formulated it as a conjecture for general d [7, Conjecture 3.6 and Theorem 3.7]. The quartic case is the first one not covered by their argument. Their conjecture is phrased for $a \geq 0$. At $a = 0$, however, the planar moments are no longer absolutely convergent and an oscillatory or regularized interpretation

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must be specified. We work throughout in the canonical absolutely convergent range $a > 0$, where the result below proves the $a > 0$ part of the $d = 4$ assertion. We make no claim about an endpoint interpretation at $a = 0$.

Our result is the following.

Theorem 1.1. *For every $a > 0$, the initial values (1.3)–(1.4) extend uniquely by (1.2) to a sequence defined for every $n \geq 0$. This sequence satisfies*

$$v_n(a) > 0 \qquad \text{for every } n \geq 0.$$

Here and throughout, an extension means a sequence of finite real numbers satisfying (1.2) at every index. Thus the uniqueness assertion concerns the prescribed Bessel initial data; it is not a classification of positive sequences arising from other initial data.

The main point is an integral sign mechanism that applies uniformly to all indices. The Bessel expression (1.4) has a positive Laplace representation. Adding independent source variables for z^4 and $\bar{z}^4$ produces four positive integral kernels, one for each residue class modulo 4. Their mixed derivative matrices are reduced by triangular row operations to

$$\left[\int_0^\infty s^p \left(\frac{s^2}{1+s^2}\right)^q \, \mathrm{d}\mu(s)\right]_{p,q=0}^{k-1}.$$

Both s and $s^2/(1+s^2)$ are strictly increasing. Andréief's identity therefore expresses the determinant as an integral of a product of two Vandermonde determinants with a fixed positive sign. This proves the required sign pattern for every finite block at once. The monotone-Vandermonde determinant argument belongs to the broader theory of sign regularity and total positivity; for the terminology and general framework, see [11, Chapter 2]. Related nonsymmetric moment determinants occur in biorthogonal ensembles [3]. Those standard mechanisms do not by themselves supply a positive measure for the oscillatory quartic Gram blocks. Indeed, a single radial source records only the total quartic degree, whereas the block entry indexed by (p, q) requires the powers of z^4 and $\bar{z}^4$ to be varied independently. The new ingredient in our proof is the two-source identity in Section 4. It first separates those two powers on truncated disks and then, after the sources are set equal, converts each of the four blocks into the positive bi-moment family of Section 3. Thus the standard determinant argument applies only after the new source representation has removed the oscillatory obstruction. The closure into four positive kernels uses the special $K_{1/2}$ evaluation of the quartic Bessel amplitude; consequently the same construction does not automatically extend to higher d. Keeping track of the triangular factors also yields an explicit positive multiple-integral formula for every leading block determinant.

The paper is organized as follows. Section 2 recalls the quartic Hermitian form and its relation to the recurrence. Section 3 proves the leading-determinant sign lemma. Section 4 derives the four positive kernels and applies the lemma to the four symmetry blocks. Section 5 then identifies the positive norm ratios with the solution of (1.2).

2. The quartic Gram form

Fix $a > 0$ and write

$$\rho_a(z) = \exp\left(-a|z|^2 + \frac{\mathrm{i}}{4}(z^4 + \bar{z}^4)\right). \tag{2.1}$$

The Gaussian factor makes all polynomial moments absolutely convergent. Let $\eta = \mathrm{e}^{\pi \mathrm{i}/4}$. Decompose the polynomial ring into C_4-weight spaces

$$\mathbb{C}[z] = \bigoplus_{r=0}^{3} \mathcal{P}_r, \qquad \mathcal{P}_r = \{f \in \mathbb{C}[z] : f(\mathrm{i}z) = \mathrm{i}^r f(z)\}.$$

For $f \in \mathcal{P}_r$, set

$$(\sigma f)(z) = (-\eta)^{-r} f(\eta z),$$

and extend σ linearly to $\mathbb{C}[z]$. With $\mathrm{d}A = \mathrm{d}(\operatorname{Re} z)\,\mathrm{d}(\operatorname{Im} z)$, define

$$(f, g)_a = \int_{\mathbb{C}} f(z)\overline{\sigma g(z)}\rho_a(z)\,\mathrm{d}A(z). \tag{2.2}$$

We use the convention that $(\ ,\ )_a$ is linear in its first argument and conjugate-linear in its second. This form is Hermitian. Indeed, distinct weight spaces are orthogonal because the density is invariant under $z \mapsto \mathrm{i}z$. For monomials in a common weight space, put

$$M_{pq}^{(r)} = \int_{\mathbb{C}} z^{r+4p}\bar{z}^{r+4q}\rho_a(z)\,\mathrm{d}A(z).$$

Since $\rho_a(\eta z) = \overline{\rho_a(z)}$, the change of variables $z \mapsto \eta z$ gives

$$\overline{M_{qp}^{(r)}} = (-1)^{q-p} M_{pq}^{(r)}.$$

Together with the twist signs below, this shows $(z^{r+4p}, z^{r+4q})_a = \overline{(z^{r+4q}, z^{r+4p})_a}$, and hence proves the assertion by sesquilinearity. If $m = r + 4q$, then

$$\sigma z^m = (-\eta)^{-r}\eta^m z^m = (-1)^{r+q} z^m. \tag{2.3}$$

Thus the Gram matrix on $\mathcal{P}_r$, in the ordered monomial basis $z^r, z^{r+4}, \ldots$, has entries

$$G_{pq}^{(r)} = (-1)^{r+q}\int_{\mathbb{C}} z^{r+4p}\bar{z}^{r+4q}\rho_a(z)\,\mathrm{d}A(z), \qquad p, q \geq 0. \tag{2.4}$$

Assume temporarily that the leading principal minors in all four blocks are nonzero. Since distinct weight spaces are orthogonal, grouping the monomials $1, z, \ldots, z^{n-1}$ by residue modulo 4 makes their Gram matrix block diagonal, with blocks that are leading principal submatrices of the four matrices (2.4); the assumption therefore makes every such Gram matrix nondegenerate, and Gram–Schmidt gives monic polynomials $P_n(z) = z^n + \cdots$. Moreover, orthogonalizing z^n against $1, \ldots, z^{n-1}$ subtracts components only from the weight space of z^n, so $P_n \in \mathcal{P}_{n \bmod 4}$. For $\ell \in \mathbb{Z}$, define

$$\tau(\ell) = \begin{cases} -1, & \ell \equiv 3 \pmod 4, \\ 1, & \ell \not\equiv 3 \pmod 4, \end{cases} \qquad \varrho(n) = (-1)^n \prod_{\ell=0}^{n-1} \tau(\ell) = (-1)^{n+\lfloor n/4 \rfloor}. \tag{2.5}$$

The signed norms $\mathfrak{h}_n$ are fixed by

$$(P_n, P_n)_a = \varrho(n)\mathfrak{h}_n. \tag{2.6}$$

Thus the scalar function $h(a)$ in (1.4) will become the first signed norm $\mathfrak{h}_0$. Section 4 will prove simultaneously that all the required minors are nonzero and that $\mathfrak{h}_n > 0$.

The pairing (2.2) is precisely the monomial-potential Hermitian product of Felder and Hoppe for $d = 4$. Indeed, $\eta = \mathrm{e}^{\pi\mathrm{i}/4}$ satisfies $\eta^4 = -1$; the density (2.1) is the density of [7, Section 3.1]; the twist σ is the map of [7, Proposition 3.1] for the right inverse χ chosen there, which is given by $\chi(r) = r$ for $0 \leq r \leq 3$; the sign τ in (2.5) is the twist sign of [7, Lemma 3.4]; and (2.6) is their normalization of the signed norms, which they denote h_n. Consequently the Gram matrices appearing in [7, Conjecture 3.6] are exactly the matrices $[(z^p, z^q)_a]_{p,q=0}^{N}$ considered below.

3. A leading-determinant sign lemma

We isolate the integral argument that drives the proof. Let $y > 0$, $\alpha \geq 0$, and $\beta \geq 0$, and define

$$\Phi_{\alpha,\beta}(w;y) = \int_0^\infty \mathrm{e}^{-ys} s^{\alpha-1/2}(1+ws^2)^{-\beta-1/2}\,\mathrm{d}s, \qquad w > 0. \tag{3.1}$$

For $p, q \geq 0$, put

$$D_{pq}^{\alpha,\beta} = \partial_u^p \partial_v^q \Phi_{\alpha,\beta}(uv;y)\big|_{u=v=1}. \tag{3.2}$$

Both $\Phi_{\alpha,\beta}$ and $D_{pq}^{\alpha,\beta}$ depend on the parameter y, which is fixed throughout; the dependence is suppressed from the notation.

Lemma 3.1 (signs of the leading source determinants)**.** *For every $k \geq 1$,*

$$\operatorname{sgn}\det\left[D_{pq}^{\alpha,\beta}\right]_{p,q=0}^{k-1} = (-1)^{k(k-1)/2}. \tag{3.3}$$

Proof. Set

$$b = \beta + \frac{1}{2}, \qquad \gamma = \frac{\alpha}{2} + \frac{1}{4}, \qquad L(y) = \Phi_{\alpha,\beta}(1;y).$$

The substitution $t = \sqrt{w}\,s$ in (3.1) gives

$$\Phi_{\alpha,\beta}(w;y) = w^{-\gamma} L\left(\frac{y}{\sqrt{w}}\right). \tag{3.4}$$

Let $\vartheta = w\partial_w$, and use the falling-factorial notation

$$(\vartheta)^{\underline{p}} = \vartheta(\vartheta-1)\cdots(\vartheta-p+1).$$

Under (3.4), ϑ acts on L as

$$\mathcal{A} = -\gamma - \frac{y}{2}\partial_y.$$

More precisely, for every smooth function g on $(0,\infty)$, a direct computation gives

$$\vartheta\left[w^{-\gamma} g\left(yw^{-1/2}\right)\right] = w^{-\gamma}(\mathcal{A}g)\left(yw^{-1/2}\right),$$

with $\mathcal{A}$ applied in the argument of g. The right-hand side has the same scaling form as the left, so the identity can be iterated for any polynomial in ϑ, and evaluation at $w = 1$ replaces the argument by y. Since $w = uv$,

$$\partial_u^p \partial_v^q F(uv)\big|_{u=v=1} = (\vartheta)^{\underline{p}}(\vartheta)^{\underline{q}} F(w)\big|_{w=1}.$$

It follows that

$$D_{pq}^{\alpha,\beta} = (\mathcal{A})^{\underline{p}}(\mathcal{A})^{\underline{q}} L(y). \tag{3.5}$$

Define

$$f(s) = \frac{s^2}{1+s^2}$$

and the bi-moments

$$B_{pq} = \int_0^\infty s^p f(s)^q \mathrm{e}^{-ys} s^{\alpha-1/2}(1+s^2)^{-\beta-1/2}\,\mathrm{d}s. \tag{3.6}$$

Each w-derivative of the integrand in (3.1) multiplies it by a factor that is bounded uniformly for w in compact subsets of $(0,\infty)$, and each y-derivative multiplies it by $-s$; the differentiated integrands are therefore dominated, locally uniformly in (w,y), by integrable functions of s. This

justifies every differentiation under the integral sign in this proof. Since $(\vartheta)^{\underline{q}} = w^q \partial_w^q$, direct differentiation of (3.1) with respect to w gives

$$(\mathcal{A})^{\underline{q}} L(y) = (-1)^q (b)_q B_{0q}, \tag{3.7}$$

where $(b)_q$ is the rising Pochhammer symbol. On the other hand,

$$(\mathcal{A})^{\underline{p}} = \sum_{i=0}^{p} c_{pi}(y) \partial_y^i, \qquad c_{pp}(y) = \left(-\frac{y}{2}\right)^p. \tag{3.8}$$

Since $\partial_y^i B_{0q} = (-1)^i B_{iq}$, (3.5)–(3.8) give a factorization

$$D = RB\Lambda, \tag{3.9}$$

because, entrywise,

$$D_{pq}^{\alpha,\beta} = (-1)^q (b)_q \sum_{i=0}^{p} (-1)^i c_{pi}(y) B_{iq}. \tag{3.10}$$

Here and below the factorization is for the $k \times k$ truncations of D and B, where R is lower triangular and Λ is diagonal, with entries

$$R_{pi} = \begin{cases} (-1)^i c_{pi}(y), & 0 \le i \le p, \\ 0, & i > p, \end{cases} \qquad \Lambda_{qq} = (-1)^q (b)_q.$$

Since $b = \beta + \frac{1}{2} > 0$, every factor $(b)_q$ is strictly positive. In particular, the only signs on the diagonal of Λ are the displayed factors $(-1)^q$. Also,

$$R_{pp} = \left(\frac{y}{2}\right)^p > 0.$$

It remains to determine the sign of $\det B$. Andréief's integration identity [1, 8] applies because $\alpha \ge 0$ gives integrability at the origin and $y > 0$ gives exponential decay at infinity, so every moment in (3.6) is finite. It gives

$$\begin{aligned} \det[B_{pq}]_{p,q=0}^{k-1} = \frac{1}{k!} \int_{(0,\infty)^k} & \det[s_j^p]_{p=0,\dots,k-1}^{j=1,\dots,k} \\ & \times \det[f(s_j)^q]_{q=0,\dots,k-1}^{j=1,\dots,k} \prod_{j=1}^{k} \mathrm{d}\mu(s_j), \end{aligned} \tag{3.11}$$

where

$$\mathrm{d}\mu(s) = \mathrm{e}^{-ys} s^{\alpha-1/2} (1+s^2)^{-\beta-1/2} \, \mathrm{d}s.$$

The two determinants in (3.11) are Vandermonde determinants. Because both s and $f(s) = s^2/(1+s^2)$ are strictly increasing on $(0,\infty)$, their product is nonnegative and is strictly positive whenever the s_j's are distinct. Moreover, μ has a strictly positive density on $(0,\infty)$, while the diagonals $s_i = s_j$ have $\mu^{\otimes k}$-measure zero. Hence $\det B > 0$.

The determinant of R is positive, whereas

$$\operatorname{sgn} \det \Lambda = (-1)^{0+1+\cdots+(k-1)} = (-1)^{k(k-1)/2}.$$

The factorization (3.9) proves (3.3). □

4. The four positive kernels

We first turn the Bessel initial function into a positive integral. For $x > 0$, the connection formula

$$J_{-1/4}(x) = \frac{J_{1/4}(x) - Y_{1/4}(x)}{\sqrt{2}}$$

reduces (1.4) to

$$h(a) = \frac{\pi^2 a}{8}\left[J_{1/4}\left(\frac{a^2}{4}\right)^2 + Y_{1/4}\left(\frac{a^2}{4}\right)^2\right]. \tag{4.1}$$

The Dixon–Ferrar formula [6] (see also [9, entry 6.518]) states that, for $x > 0$ and $|\operatorname{Re}\nu| < \frac{1}{2}$,

$$J_\nu(x)^2 + Y_\nu(x)^2 = \frac{8\cos(\pi\nu)}{\pi^2}\int_0^\infty K_{2\nu}(2x\sinh t)\,\mathrm{d}t. \tag{4.2}$$

Using

$$K_{1/2}(u) = \sqrt{\frac{\pi}{2u}}\mathrm{e}^{-u}$$

in (4.1), and then setting $s = \sinh t$, gives

$$h(a) = \sqrt{\frac{\pi}{2}}\int_0^\infty \frac{\mathrm{e}^{-a^2 s/2}}{\sqrt{s(1+s^2)}}\,\mathrm{d}s. \tag{4.3}$$

In particular, differentiation under this positive integral gives

$$h(a) > 0, \qquad h'(a) = -a\sqrt{\frac{\pi}{2}}\int_0^\infty \frac{\mathrm{e}^{-a^2 s/2}\sqrt{s}}{\sqrt{1+s^2}}\,\mathrm{d}s < 0. \tag{4.4}$$

Thus both denominators in the prescribed initial values (1.3) are nonzero for $a > 0$. Positivity of h'', and hence positivity of v_1, will follow from the third kernel in Lemma 4.3.

Set

$$y = \frac{a^2}{2}, \qquad C = \sqrt{\frac{\pi}{2}}.$$

We now introduce an angular source. For $u, v > 0$, define the radial generating function

$$\mathcal{Z}(a; u, v) = \pi\int_0^\infty \mathrm{e}^{-at} J_0\left(\frac{t^2\sqrt{uv}}{2}\right)\,\mathrm{d}t. \tag{4.5}$$

This is the angularly integrated two-source expression. It is not being defined as a two-dimensional integral away from $u = v$; such an integral need not be absolutely convergent. For each fixed R, however, termwise angular integration is legitimate and gives

$$\int_0^{2\pi} \exp\left(\frac{\mathrm{i}R^4}{4}(u\mathrm{e}^{4\mathrm{i}\theta} + v\mathrm{e}^{-4\mathrm{i}\theta})\right)\,\mathrm{d}\theta = 2\pi J_0\left(\frac{R^4\sqrt{uv}}{2}\right), \tag{4.6}$$

and $t = R^2$ gives (4.5). On compact subsets of $(0,\infty)^2$, every u, v-derivative of the Bessel integrand is bounded by e^{-at} times a polynomial in t: every derivative of J_0 is a finite linear combination of integer-order Bessel functions, which are bounded on the real axis, and the chain-rule factors involving u and v are uniformly bounded on such a compact set. Thus derivatives may be passed under the t-integral. The precise comparison with the planar moments is recorded in Lemma 4.2 below.

Lemma 4.1 (source normalization)**.** *For every $a > 0$,*

$$\int_{\mathbb{C}} \rho_a(z)\,\mathrm{d}A(z) = \pi \int_0^\infty \mathrm{e}^{-at} J_0\left(\frac{t^2}{2}\right)\mathrm{d}t = \mathcal{Z}(a;1,1) = h(a), \tag{4.7}$$

where h is the Bessel combination (1.4)*.*

Proof. Polar coordinates and (4.6), with $u = v = 1$, give the first two equalities in (4.7). It remains to identify the radial Laplace integral with the Bessel combination, including its normalization. Put

$$Z(a) = \pi \int_0^\infty \mathrm{e}^{-at} J_0(t^2/2)\,\mathrm{d}t, \qquad g(t) = J_0(t^2/2).$$

The Bessel equation for J_0 gives

$$g''(t) + \frac{1}{t}g'(t) + t^2 g(t) = 0, \qquad \text{or equivalently} \qquad -t^3 g(t) = (tg'(t))'.$$

At the origin, the power series for J_0 gives

$$g(t) = 1 + O(t^4), \qquad g'(t) = O(t^3),$$

so in particular $tg'(t) \to 0$. At infinity, g and g' have at most polynomial growth, while e^{-at} gives exponential decay. These estimates justify the following differentiations and ensure that all boundary terms in the two integrations by parts vanish. More explicitly,

$$\int_0^\infty \mathrm{e}^{-at}(tg'(t))'\,\mathrm{d}t = a\int_0^\infty \mathrm{e}^{-at} tg'(t)\,\mathrm{d}t = a\int_0^\infty \mathrm{e}^{-at}(at-1)g(t)\,\mathrm{d}t.$$

We therefore obtain

$$\begin{aligned} Z'''(a) + a^2 Z'(a) + aZ(a) &= \pi \int_0^\infty \mathrm{e}^{-at}(-t^3 - a^2 t + a)g(t)\,\mathrm{d}t \\ &= 0. \end{aligned}$$

Moreover, after setting $u = at$, boundedness of J_0 on the real axis and dominated convergence yield

$$\lim_{a\to\infty} aZ(a) = \pi \int_0^\infty \mathrm{e}^{-u}\,\mathrm{d}u = \pi. \tag{4.8}$$

These two properties determine Z uniquely within the three-dimensional solution space: for any solution H, the existence of a finite limit of $aH(a)$ eliminates the two independent oscillatory combinations, and the value of that limit fixes the remaining coefficient. To verify this explicitly, let Q be a product of two solutions of the Bessel equation of order ν. Its symmetric-square equation is

$$x^2 Q''' + 3xQ'' + (4x^2 + 1 - 4\nu^2)Q' + 4xQ = 0.$$

With $x = a^2/4$ and $\nu = 1/4$, direct substitution shows that $H(a) = aQ(a^2/4)$ satisfies

$$H''' + a^2 H' + aH = 0$$

and hence that the solution space of this equation is spanned by

$$aJ_\nu(x)^2, \qquad aJ_\nu(x)Y_\nu(x), \qquad aY_\nu(x)^2.$$

These three functions are linearly independent because, on any interval avoiding the zeros of Y_ν, a linear relation among them would be a quadratic polynomial relation in the function J_ν/Y_ν; the latter is nonconstant there, since the Wronskian identity [12, equation (10.5.2)] gives $(J_\nu/Y_\nu)' = -2/(\pi x Y_\nu^2) \neq 0$. Since the equation has order three, they form a basis of its

solution space. With $\chi = x - \nu\pi/2 - \pi/4$, the standard large-$x$ Bessel asymptotics [12, §10.17, equations (10.17.3)–(10.17.4)] give

$$\begin{aligned} a\big[aJ_\nu(x)^2\big] &= \frac{4}{\pi}(1+\cos(2\chi)) + O(a^{-2}), \\ a\big[aJ_\nu(x)Y_\nu(x)\big] &= \frac{4}{\pi}\sin(2\chi) + O(a^{-2}), \\ a\big[aY_\nu(x)^2\big] &= \frac{4}{\pi}(1-\cos(2\chi)) + O(a^{-2}). \end{aligned}$$

Write a general solution as

$$H = c_1 aJ_\nu(x)^2 + c_2 aJ_\nu(x)Y_\nu(x) + c_3 aY_\nu(x)^2.$$

Since $2\chi = a^2/2 - 3\pi/4$ is continuous, strictly increasing, and unbounded, there are sequences tending to infinity on which its value modulo 2π is respectively $0, \pi/2, \pi, 3\pi/2$. Comparing the limit of $aH(a)$ along these four sequences shows that $c_1 - c_3 = 0$ and $c_2 = 0$. The condition $\lim_{a\to\infty} aH(a) = \pi$ then gives $c_1 = c_3 = \pi^2/8$. Thus the unique solution with the prescribed limit is

$$H(a) = \frac{\pi^2 a}{8}\left[J_{1/4}\left(\frac{a^2}{4}\right)^2 + Y_{1/4}\left(\frac{a^2}{4}\right)^2\right].$$

This is $h(a)$ by (4.1), proving the final equality in (4.7). The radial representation and its $d = 4$ Bessel evaluation also appear in [7, pp. 10–11]. □

Put $w = uv$. The change of variables $t = w^{-1/4}x$ in (4.5), followed by (4.3), now gives

$$\mathcal{Z}(a;u,v) = w^{-1/4}h(aw^{-1/4}),$$

or, more explicitly,

$$\mathcal{Z}(a;u,v) = Cw^{-1/4}\int_0^\infty \frac{\mathrm{e}^{-yw^{-1/2}t}}{\sqrt{t(1+t^2)}}\,\mathrm{d}t.$$

The further substitution $t = \sqrt{w}\,s$ therefore gives

$$\mathcal{Z}(a;u,v) = C\int_0^\infty \frac{\mathrm{e}^{-ys}}{\sqrt{s(1+uvs^2)}}\,\mathrm{d}s = C\Phi_{0,0}(uv;y). \tag{4.9}$$

For $r = 0, 1, 2, 3$, put

$$\mathcal{Z}_r = (-\partial_a)^r\mathcal{Z}.$$

Lemma 4.2 (source derivatives and planar moments)**.** *For $r \in \{0,1,2,3\}$ and $p, q \geq 0$,*

$$\partial_u^p\partial_v^q\mathcal{Z}_r(a;u,v)|_{u=v=1} = \left(\frac{\mathrm{i}}{4}\right)^{p+q}\int_{\mathbb{C}} z^{r+4p}\bar{z}^{r+4q}\rho_a(z)\,\mathrm{d}A(z). \tag{4.10}$$

Proof. Fix a compact rectangle $Q \subset (0,\infty)^2$ containing $(1,1)$. Repeated differentiation of the Bessel integrand in (4.5), using the standard derivative recurrences for J_ν, gives, for fixed p, q, r, a bound of the form

$$\left|\partial_u^p\partial_v^q(-\partial_a)^r\left[\mathrm{e}^{-at}J_0\left(\frac{t^2\sqrt{uv}}{2}\right)\right]\right| \leq C_{Q,p,q,r}\mathrm{e}^{-at}(1+t)^{2p+2q+r}.$$

Thus dominated convergence permits all the indicated derivatives to pass through the t-integral.

For $R > 0$, define the truncated radial source

$$\mathcal{Z}^{[R]}(a;u,v) = \pi\int_0^{R^2}\mathrm{e}^{-at}J_0\left(\frac{t^2\sqrt{uv}}{2}\right)\mathrm{d}t, \qquad \mathcal{Z}_r^{[R]} = (-\partial_a)^r\mathcal{Z}^{[R]}.$$

For real $u, v > 0$, also put

$$\mathcal{I}^{[R]}(a;u,v) = \int_{|z|\le R} \exp\left(-a|z|^2 + \frac{\mathrm{i}}{4}(uz^4 + v\bar{z}^4)\right) \mathrm{d}A(z).$$

Applying (4.6) in polar coordinates gives $\mathcal{I}^{[R]} = \mathcal{Z}^{[R]}$. All u, v-derivatives here are real parameter derivatives, and all differentiations are over the compact disk. At $u = v = 1$, each u-derivative contributes $\mathrm{i}z^4/4$, each v-derivative contributes $\mathrm{i}\bar{z}^4/4$, and each $-\partial_a$ contributes $|z|^2 = z\bar{z}$. Therefore

$$\partial_u^p \partial_v^q \mathcal{Z}_r^{[R]}(a;u,v)\Big|_{u=v=1} = \left(\frac{\mathrm{i}}{4}\right)^{p+q} \int_{|z|\le R} z^{r+4p}\bar{z}^{r+4q}\rho_a(z)\,\mathrm{d}A(z).$$

On the radial side, the preceding integrable bound and dominated convergence show that the left-hand side converges to the left-hand side of (4.10) as $R \to \infty$. On the planar side, the integrand is bounded in absolute value by $|z|^{2r+4p+4q}\mathrm{e}^{-a|z|^2}$, so dominated convergence gives the full planar moment. This proves (4.10). □

Lemma 4.3. *The four radial source functions are*

$$\begin{array}{c|c} r & \mathcal{Z}_r(a;u,v) \\ \hline 0 & C\Phi_{0,0}(uv;y), \\ 1 & aC\Phi_{1,0}(uv;y), \\ 2 & 2C\Phi_{1,1}(uv;y), \\ 3 & 2aC\Phi_{2,1}(uv;y). \end{array} \tag{4.11}$$

Proof. The first two rows follow by differentiating (4.9). For the third row, integration by parts gives

$$\begin{aligned} y\int_0^\infty \frac{\mathrm{e}^{-ys}s^{3/2}}{\sqrt{1+uvs^2}}\,\mathrm{d}s &= \frac{1}{2}\int_0^\infty \frac{\mathrm{e}^{-ys}s^{1/2}}{\sqrt{1+uvs^2}}\,\mathrm{d}s \\ &\quad + \int_0^\infty \frac{\mathrm{e}^{-ys}s^{1/2}}{(1+uvs^2)^{3/2}}\,\mathrm{d}s. \end{aligned} \tag{4.12}$$

The boundary terms vanish at 0 and ∞. Since

$$(-\partial_a)^2 = \partial_y + 2y\partial_y^2,$$

equation (4.12) gives $\mathcal{Z}_2 = 2C\Phi_{1,1}$. Since

$$\partial_y\Phi_{1,1}(uv;y) = -\Phi_{2,1}(uv;y), \qquad -\partial_a = -a\partial_y,$$

one more differentiation gives

$$\mathcal{Z}_3 = -\partial_a\mathcal{Z}_2 = 2aC\Phi_{2,1}(uv;y).$$

In particular, at $u = v = 1$,

$$h''(a) = \mathcal{Z}_2(a;1,1) = 2C\Phi_{1,1}(1;y) > 0.$$

□

For $k \ge 1$, set

$$\Delta_k^{(r)} = \det[G_{pq}^{(r)}]_{p,q=0}^{k-1}, \qquad \Delta_0^{(r)} = 1. \tag{4.13}$$

Proposition 4.4. *For every $r \in \{0,1,2,3\}$ and $k \ge 1$,*

$$\operatorname{sgn}\Delta_k^{(r)} = (-1)^{rk+k(k-1)/2}. \tag{4.14}$$

Proof. Let

$$D_{pq}^{(r)} = \partial_u^p \partial_v^q \mathcal{Z}_r(a;u,v)|_{u=v=1}.$$

Every scalar multiplier in (4.11) is positive. Lemma 3.1, applied to the four parameter pairs

$$(\alpha,\beta) = (0,0),(1,0),(1,1),(2,1),$$

therefore gives

$$\operatorname{sgn}\det[D_{pq}^{(r)}]_{p,q=0}^{k-1} = (-1)^{k(k-1)/2}. \tag{4.15}$$

Lemma 4.2, combined with (2.4), gives

$$G_{pq}^{(r)} = (-1)^{r+q}\left(\frac{4}{\mathrm{i}}\right)^{p+q} D_{pq}^{(r)}. \tag{4.16}$$

Let $S_k = k(k-1)/2$. Factoring the row and column phases in (4.16) yields

$$\Delta_k^{(r)} = (-1)^{rk} 16^{S_k} \det[D_{pq}^{(r)}]_{p,q=0}^{k-1}. \tag{4.17}$$

Indeed, taking $(4/\mathrm{i})^p$ from row p and $(-1)^{r+q}(4/\mathrm{i})^q$ from column q produces

$$\left(\frac{4}{\mathrm{i}}\right)^{S_k} (-1)^{rk+S_k} \left(\frac{4}{\mathrm{i}}\right)^{S_k} = (-1)^{rk} 16^{S_k},$$

because $(4/\mathrm{i})^2 = -16$. Now (4.14) follows from (4.15). For example, the first two cases give the useful checks

$$\operatorname{sgn}\Delta_1^{(r)} = (-1)^r, \qquad \operatorname{sgn}\Delta_2^{(r)} = -1.$$

□

Corollary 4.5 (positive multiple-integral formula)**.** *For $r = 0,1,2,3$, define*

r	A_r	α_r	β_r
0	C	0	0
1	aC	1	0
2	$2C$	1	1
3	$2aC$	2	1

$$\left(C = \sqrt{\frac{\pi}{2}}, \quad y = \frac{a^2}{2}\right). \tag{4.18}$$

For $k \geq 1$, if $S_k = k(k-1)/2$, then

$$\begin{aligned}(-1)^{rk+S_k}\Delta_k^{(r)} = \frac{A_r^k (4a^2)^{S_k}}{k!} \prod_{q=0}^{k-1} \left(\beta_r + \frac{1}{2}\right)_q \\ \times \int_{(0,\infty)^k} \prod_{j=1}^{k} \frac{\mathrm{e}^{-y s_j} s_j^{\alpha_r - 1/2}}{(1+s_j^2)^{\beta_r + k - 1/2}} \\ \times \prod_{1 \leq i < j \leq k} (s_j - s_i)^2 (s_i + s_j) \prod_{j=1}^{k} \mathrm{d}s_j.\end{aligned} \tag{4.19}$$

In particular, the right-hand side is strictly positive.

Proof. For the parameters in (4.18), (4.11) says that $D^{(r)} = A_r D^{\alpha_r,\beta_r}$. The triangular factorization (3.9), with $b = \beta_r + \frac{1}{2}$, gives

$$\det D^{\alpha_r,\beta_r} = (-1)^{S_k} \left(\frac{y}{2}\right)^{S_k} \prod_{q=0}^{k-1} \left(\beta_r + \frac{1}{2}\right)_q \det B.$$

Combining this identity with (4.17), and using $16^{S_k}(y/2)^{S_k} = (4a^2)^{S_k}$, fixes the prefactor in (4.19).

It remains only to rewrite the Andréief integrand. For $f(s) = s^2/(1+s^2)$,

$$f(s_j) - f(s_i) = \frac{(s_j - s_i)(s_j + s_i)}{(1+s_i^2)(1+s_j^2)}.$$

Consequently the product of the two Vandermonde determinants in (3.11) is

$$\prod_{1 \le i < j \le k} \frac{(s_j - s_i)^2 (s_i + s_j)}{(1+s_i^2)(1+s_j^2)}.$$

Each factor $1 + s_j^2$ occurs $k-1$ times in the denominator. Substitution into (3.11) proves (4.19); strict positivity follows because the integrand is positive away from the diagonals $s_i = s_j$, which have measure zero, as in the proof of Lemma 3.1. □

Corollary 4.6 (quartic Gram nondegeneracy)**.** *For every $N \ge 0$, the leading Gram matrix $\big[(z^p, z^q)_a\big]_{p,q=0}^{N}$ is nondegenerate, and the signed norms in (2.6) satisfy*

$$\mathfrak{h}_n > 0 \qquad (n \ge 0).$$

In particular, the $a > 0$ part of [7, Conjecture 3.6] *holds for $d = 4$.*

Proof. The block sign formula shows first that every $\Delta_k^{(r)}$ is nonzero. After simultaneously permuting rows and columns to group monomials by their residue modulo 4, every finite leading Gram matrix is block diagonal, and its blocks are leading matrices of the four types in (4.13). More precisely, the block of residue r has size

$$k_r(N) = \begin{cases} \lfloor (N-r)/4 \rfloor + 1, & r \le N, \\ 0, & r > N. \end{cases}$$

It is therefore nondegenerate, and all the monic orthogonal polynomials exist. Let $n = r + 4j$. The orthogonality conditions within $\mathcal{P}_r$ determine the coefficients of P_n by Cramer's rule, and pairing P_n with z^{r+4j} gives the classical minor ratio

$$(P_n, P_n)_a = \frac{\Delta_{j+1}^{(r)}}{\Delta_j^{(r)}}, \tag{4.20}$$

which uses only the nonvanishing of the leading minors, not definiteness; see [13, Section 2.2]. Taking the quotient of the signs in (4.14) gives

$$\operatorname{sgn}(P_n, P_n)_a = (-1)^{r+j}.$$

But (2.5) gives $\varrho(r+4j) = (-1)^{r+j}$. Hence $\mathfrak{h}_n = \varrho(n)(P_n, P_n)_a > 0$. By the identification at the end of Section 2, the nondegeneracy assertion is exactly the $d = 4$, $a > 0$ case of [7, Conjecture 3.6]. □

The multiple-integral formula also gives the norms and recurrence coefficients directly as determinant ratios. Namely, if $n = r + 4j$, then

$$\mathfrak{h}_n = \frac{|\Delta_{j+1}^{(r)}|}{|\Delta_j^{(r)}|}, \qquad v_n = \frac{1}{a}\frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n}, \tag{4.21}$$

where each determinant on the right is given explicitly by (4.19). Since the integrand in (4.19) is positive, this evaluation is free of cancellation, in contrast with a direct computation of the oscillatory moment determinants.

5. Identification with the quartic recurrence

It remains to verify that the positive norm ratios give precisely the sequence in Theorem 1.1. Define

$$V_n = \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n}, \qquad v_n = \frac{V_n}{a}. \tag{5.1}$$

For the recurrence formulas below, also set $V_j = 0$ for $j < 0$, in agreement with the convention $v_j = 0$ in (1.2). Corollary 4.6 already gives

$$v_n > 0 \qquad (n \geq 0). \tag{5.2}$$

For clarity, we include the recurrence calculation from the quartic Gram form; it is the specialization of [7, Lemma 3.4 and equations (3.1)–(3.3)]. If g has C_4-weight ℓ, the definition of the twist gives

$$\sigma(zg) = -\tau(\ell) z\, \sigma g. \tag{5.3}$$

Indeed, for $\ell = 0, 1, 2$ this follows directly by comparing the twist factors in weights ℓ and $\ell + 1$; for $\ell = 3$, the weight returns to 0, and $\eta^4 = -1$ supplies the exceptional sign $\tau(3) = -1$.

We use the Wirtinger derivative $\partial_z = \frac{1}{2}(\partial_x - \mathrm{i}\partial_y)$. Then $\partial_z \overline{\sigma g(z)} = 0$ and

$$\partial_z \rho_a(z) = (-a\bar{z} + \mathrm{i}z^3)\rho_a(z).$$

Differentiating (2.1) and integrating by parts in (2.2), with the boundary term removed by the Gaussian decay, therefore give

$$\begin{aligned} \big((\partial_z + \mathrm{i}z^3)f, g\big)_a &= \int_{\mathbb{C}} (\partial_z f + \mathrm{i}z^3 f)\overline{\sigma g}\, \rho_a \, \mathrm{d}A \\ &= a \int_{\mathbb{C}} f \bar{z}\, \overline{\sigma g}\, \rho_a \, \mathrm{d}A \\ &= -\tau(\ell) a(f, zg)_a, \end{aligned} \tag{5.4}$$

where the last equality uses (5.3). We spell out the bandedness consequence, since weight considerations alone do not exclude more distant lower terms. Write $L = \partial_z + \mathrm{i}z^3$. If $j \leq n - 7$ has the same weight as zP_n, then $\deg LP_j \leq j + 3 < n$, and

$$(LP_j, P_n)_a = -\tau(n) a(P_j, zP_n)_a = 0.$$

Hermitian symmetry therefore gives $(zP_n, P_j)_a = 0$. Similarly, if $j \leq n - 5$ has the same weight as LP_n, then

$$(LP_n, P_j)_a = -\tau(j) a(P_n, zP_j)_a = 0,$$

because $\deg(zP_j) < n$. Weight, degree, and orthogonality now give

$$\begin{aligned} zP_n &= P_{n+1} + \frac{\mathrm{i}}{a}\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-3}} P_{n-3}, \\ (\partial_z + \mathrm{i}z^3)P_n &= \mathrm{i}P_{n+3} + a\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}} P_{n-1}, \end{aligned} \tag{5.5}$$

where terms with negative indices are omitted. Pairing the first expansion with P_{n-3}, and using (5.4) with $f = P_{n-3}$, $g = P_n$, determines its lower coefficient. Pairing the second expansion with P_{n-1} determines the other one. In both calculations one uses

$$\varrho(n) = \tau(n)\varrho(n-3), \qquad \varrho(n) = -\tau(n-1)\varrho(n-1).$$

For completeness, if c_n is the coefficient of P_{n-3} in zP_n, then

$$\mathrm{i}\varrho(n)\mathfrak{h}_n = (LP_{n-3}, P_n)_a = -\tau(n)a(P_{n-3}, zP_n)_a,$$
$$c_n = \frac{(zP_n, P_{n-3})_a}{\varrho(n-3)\mathfrak{h}_{n-3}} = \frac{\mathrm{i}}{a}\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-3}},$$

where Hermitian symmetry and $\varrho(n) = \tau(n)\varrho(n-3)$ are used in the second line. Similarly, if d_n is the coefficient of P_{n-1} in LP_n, then

$$d_n = \frac{(LP_n, P_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}}$$
$$= -\tau(n-1)a\frac{(P_n, zP_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}} = a\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}},$$

because the leading term of zP_{n-1} is P_n and $\varrho(n) = -\tau(n-1)\varrho(n-1)$. These computations also make the phase cancellation in (5.5) explicit.

The coefficient of P_{n-1} in z^3P_n receives three contributions: the downward step in the first formula of (5.5) may occur after zero, one, or two upward steps. Hence that coefficient is

$$\frac{\mathrm{i}}{a}V_{n-1}\left(V_{n-2}V_{n-3} + V_nV_{n-2} + V_{n+1}V_n\right).$$

Subtracting $\mathrm{i}z^3P_n$ from the second formula in (5.5), and using that the coefficient of P_{n-1} in $\partial_z P_n$ is n (the basis is monic and graded, so this coefficient is the leading coefficient of $\partial_z P_n$), gives, after putting $m = n-1$,

$$V_m\left[1 + \frac{1}{a^2}\left(V_{m-1}V_{m-2} + V_{m+1}V_{m-1} + V_{m+2}V_{m+1}\right)\right] = \frac{m+1}{a}. \tag{5.6}$$

Here $n \geq 1$, hence $m \geq 0$; the convention $V_j = 0$ for $j < 0$ incorporates the omitted negative-index terms when $m = 0, 1$. For example, at $m = 0$, equation (5.6) reads

$$V_0\left(1 + \frac{V_2V_1}{a^2}\right) = \frac{1}{a},$$

and after $V_j = av_j$ this is

$$v_0(1 + v_2v_1) = a^{-2},$$

which is exactly the $n = 0$ case of (1.2). With $V_m = av_m$, equation (5.6) is (1.2); before that substitution, it is the case $d = 4$ of (1.1).

Finally, $P_0 = 1$, $P_1 = z$, and $P_2 = z^2$. From (4.7), differentiation under the absolutely convergent planar integral gives

$$h'(a) = -\int_{\mathbb{C}} |z|^2\rho_a(z)\,\mathrm{d}A(z), \qquad h''(a) = \int_{\mathbb{C}} |z|^4\rho_a(z)\,\mathrm{d}A(z).$$

The twist signs (2.3) and $\varrho(0) = -\varrho(1) = \varrho(2) = 1$ consequently give

$$\mathfrak{h}_0 = h, \qquad \mathfrak{h}_1 = -h', \qquad \mathfrak{h}_2 = h''.$$

Therefore (5.1) gives exactly (1.3). Conversely, solving (1.2) for its newest term gives

$$v_{m+2} = \frac{(m+1)a^{-2}/v_m - 1 - v_{m-1}v_{m-2} - v_{m+1}v_{m-1}}{v_{m+1}}, \qquad m \geq 0. \tag{5.7}$$

Thus v_{m+2} is uniquely determined by the preceding terms whenever $v_mv_{m+1} \neq 0$. The norm-ratio sequence (5.1) is positive, so (5.7) applies at every step and constructs a global extension of the prescribed initial data. If $(\widetilde{v}_n)$ is any other extension, then $\widetilde{v}_0 = v_0$ and $\widetilde{v}_1 = v_1$. Assuming $\widetilde{v}_j = v_j$ through $j = m+1$, the positivity of v_mv_{m+1} allows the equation at index m to be solved

for its newest term, forcing $\widetilde{v}_{m+2} = v_{m+2}$. Induction proves uniqueness. Together with (5.2), this proves Theorem 1.1.

Remark 5.1. The uniqueness in Theorem 1.1 concerns the sequence with the prescribed initial data. The argument does not claim a classification of positive solutions with other initial data, a closed evaluation of the Gram determinants, asymptotic formulas for v_n, or a result at the endpoint $a = 0$.

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